



World Fiscal Systems for *Unconventional* Oil & Gas

Designing Fiscal Terms, Contracts and Economic Models for Shale, CBM, Oil Sands and Emerging Resources



World Fiscal Systems for *Unconventional* Oil & Gas

This course provides a comprehensive overview of the fiscal terms and regulatory frameworks that govern unconventional oil and gas developments worldwide. It covers leases, concessions, licences and production sharing contracts (PSCs) as applied to shale oil and shale gas, coal bed methane (CBM), oil sands and emerging resources such as gas hydrates. Participants gain both conceptual understanding and hands-on skills in designing and analysing fiscal systems for unconventional projects in North America, Latin America, Europe, Africa, Asia and Australia.

The programme starts with the geological and technical characteristics of unconventional resources—how they differ from conventional fields in terms of well performance, decline behaviour, drilling intensity and environmental footprint—and why these differences require distinct fiscal and regulatory approaches. It then examines the range of fiscal instruments used across key jurisdictions, including royalties, bonuses, rental payments, corporate income tax, profit-based levies, resource extraction taxes and PSC profit-oil structures, highlighting how these are adapted to unconventional projects.

A central feature of the course is an advanced Excel-based economic and fiscal model that simulates typical shale, CBM and oil sands projects under alternative fiscal regimes. Using this interactive tool, participants assess project cash flows, government take, investor returns and risk under different price, cost and production scenarios. Case studies and exercises show how to modify fiscal terms to attract investment, share resource wealth fairly, and align unconventional development with climate and ESG objectives.



LOCATION

CLASSROOM & VIRTUAL DELIVERY

LONDON PETRO ACADEMY
Empowering Professionals for a Sustainable Future



Course Objectives

By the end of the course, participants will be able to:

- Describe the main fiscal instruments and contract types used for unconventional oil and gas projects, including leases, concessions, licences and PSCs.
- Explain the geological, technical and economic characteristics of shale oil, shale gas, CBM, oil sands and gas hydrates, and how these differ from conventional developments.
- Analyse the cash-flow profiles, profitability and risk characteristics of unconventional projects in key regions such as North America, Latin America, Europe, Africa and Asia.
- Interpret and compare fiscal systems for unconventional resources in countries including the USA, Canada, Argentina, Brazil, UK, Germany, Poland, Russia, Algeria, Indonesia, China, India and Australia.
- Use an Excel-based economic model to simulate unconventional projects, calculate investor returns, government revenues, government take and risk-sharing metrics.
- Design and adjust fiscal terms (royalties, taxes, profit-oil shares, state participation, bonuses, rental payments and PSC parameters) to encourage investment while protecting government interests.
- Incorporate ESG, environmental and climate-policy considerations into unconventional fiscal design, including emissions, water use, community impacts and carbon pricing.
- Evaluate resource-wealth-sharing mechanisms and progressive fiscal features based on volume, price and costs for unconventional developments.
- Assess the timing and stability of government take (front-end vs back-end loading) and understand how to manage fiscal stability and contract change risk.
- Prepare and critique fiscal alternatives and negotiation positions for unconventional licences and PSCs.



This course is designed for professionals involved in the design, negotiation, administration or analysis of fiscal regimes and contracts for unconventional oil and gas. It is suitable for both government and industry participants, including:

- Government officials and regulators
- CEOs, CFOs and country managers
- Lawyers and legal advisers working on upstream contracts
- Economists and economic analysts
- Engineers, geologists and geophysicists involved in unconventional projects
- Tax advisers and accountants
- Negotiators and commercial advisers
- Contract and licence administrators
- Economic and fiscal planners
- Policy makers and energy-transition strategists

It is particularly valuable for those who already understand conventional fiscal systems and need to extend their expertise to unconventional resources, as well as teams developing new unconventional policies or renegotiating existing terms.

Day 1 – Unconventional Resources and North American Fiscal Systems

Course Introduction and Software Orientation

- Overview of course structure and objectives
- Introduction to the Excel-based economic and fiscal model
- Inputs, outputs, price and cost sensitivity analysis

Shale Oil & Shale Gas – Geology, Technology and Environment

- Shale oil and shale gas geology and resource characteristics
- Definitions: shale oil vs oil shale; tight oil vs conventional
- Production technology: horizontal drilling, multi-stage fracturing, pad drilling
- World resources, key plays and development trends
- Environmental and ESG issues: water use, induced seismicity, methane emissions, community impacts

North American Shale Plays – Economics and Fiscal Terms

- Typical well economics and decline profiles for shale oil and gas
- Overview of major North American shale plays (e.g. Permian, Bakken, Marcellus, Montney)
- Canadian fiscal systems: bonuses, rentals, royalties (Alberta shale gas, British Columbia NPI, Saskatchewan)
- US fiscal systems: federal and state royalties; production and severance taxes; property taxes

Corporate Income Tax and Fiscal Features

- Corporate income tax basics for unconventional projects
- Ring-fencing, consolidation and loss carry-forward
- Treatment of government payments, bonuses, operating and capital costs
- Depletion allowances, uplifts and other incentives

Case Study and Modelling Exercise

- Economic analysis of selected Canadian and US shale plays
- Simulation of project cash flows and government take under alternative fiscal structures



Day 2 – CBM, Oil Sands and International Shale Terms

Coal Bed Methane (CBM) – Geology, Technology and Fiscal Terms

CBM geology, reservoir behaviour and production technology

Global CBM resources and development status

North American and Australian CBM fiscal terms (Alberta, British Columbia, Nova Scotia, Queensland)

Environmental and ESG issues for CBM

Oil Sands – Economics and Fiscal Frameworks

Geology, technology and environmental challenges of oil sands

Typical SAGD and mining project economics

Alberta oil sands royalties and special features (FTP, price caps, deemed interest)

Interaction with carbon and climate policies

International Shale Plays – Europe, Latin America and Russia

Economics and regulatory issues for international shale plays

European fiscal terms: UK, Germany, France, Spain, Bulgaria, Poland; oil shales in Estonia

Latin American fiscal terms: Colombia and Brazil; updated context on Argentina's shale developments and Gas Plus programme

Russian fiscal terms for shale and unconventional oil: mineral resource extraction tax, export duties and special incentives

Joint Ventures, Carried Interests and State Participation

Joint Operating Agreements and farm-out structures

Carried interest provisions and participation options

State company participation and its impact on economics and risk

Treatment of JV cash flows under corporate income tax

Case Study and Modelling Exercise

Comparative analysis of international shale and CBM projects under different fiscal regimes

Impact of JV and state participation structures on government take and investor returns



Day 3 – Production Sharing, Fiscal Design and Resource Wealth Sharing

Production Sharing Contracts for Unconventional Resources

Classic PSC structures (e.g. Indonesia model)

Profit oil: fixed and sliding scales

Cost oil definitions, limits and recovery rules

Domestic market obligations and local supply requirements

Application to unconventional oil and gas (shale, CBM, oil sands) in countries such as Indonesia, China and India

Advanced Fiscal Design for Unconventional Resources

Global trends in special terms for unconventional projects

Differences between gas-focused and oil-focused fiscal designs

Encouraging re-investment and continuous development in unconventional plays

Integrating carbon pricing, emissions regulations and ESG requirements into fiscal structures

Resource Wealth Sharing and Fiscal Health

Methods for sharing resource wealth between government and investors

Progressive features linked to production volume, price and cost levels

Encouraging maximum resource recovery and cost efficiency

Avoiding gold-plating and distortions; lessons from past systems (e.g. India, Algeria)

Timing, Stability and Level of Government Take

Front-end vs back-end loading of government revenues

Rating levels of government take for shale oil, shale gas and CBM

Fiscal stability provisions and renegotiation risks

Strategies for maintaining investment attractiveness while adjusting terms over time

Design Workshop and Final Modelling Exercise

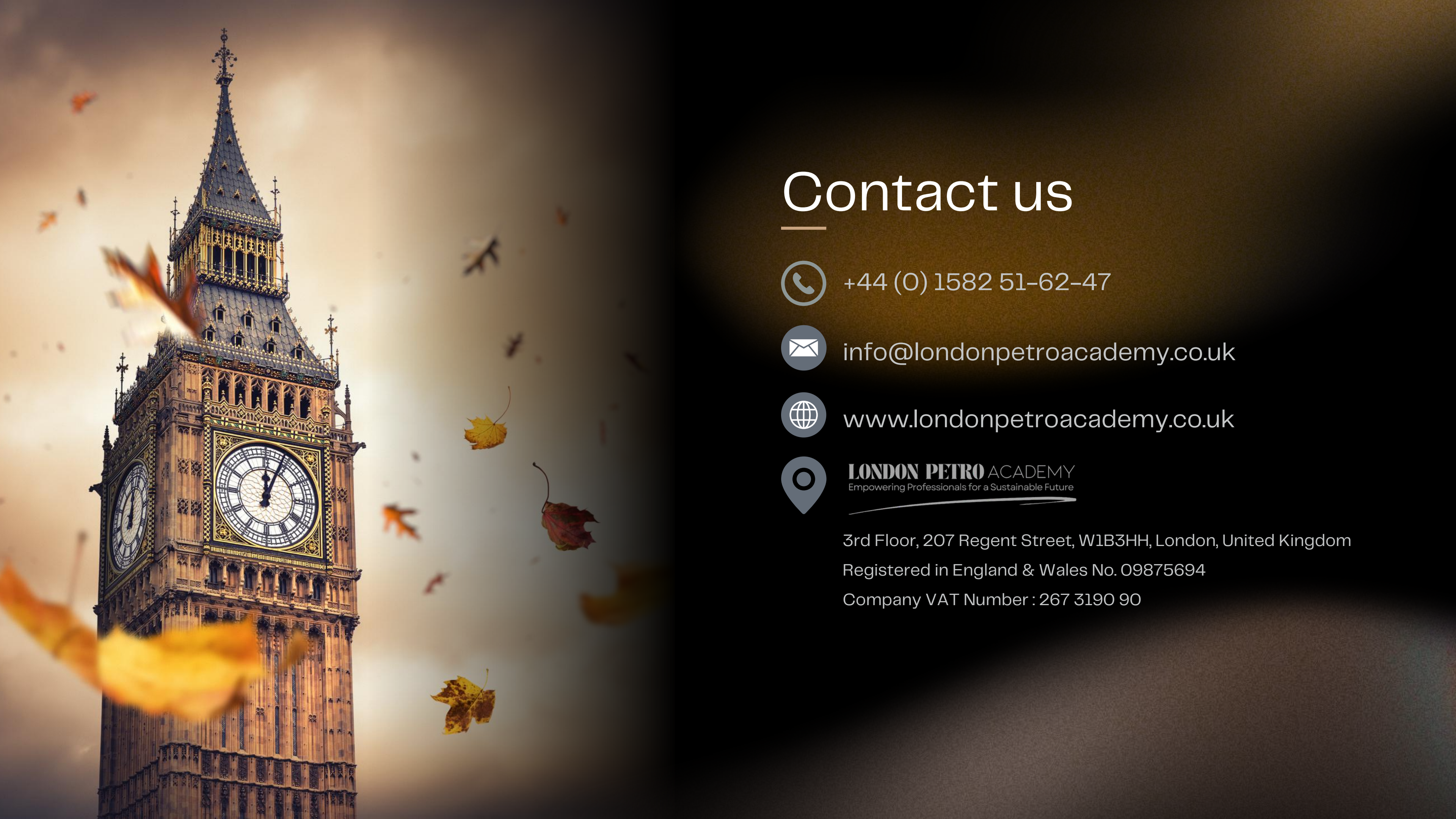
Group exercise: modifying a conventional PSC to make it suitable for shale oil or shale gas

Using the model to test alternative terms and assess project viability and government take

Price and cost sensitivity analysis; risk-sharing and incentive indices

Discussion of key findings and best-practice features for unconventional fiscal systems





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